ANALYTICAL CUMULATIVE EXAMINATION
KINETIC METHODS
October 27, 1990

1. The reaction between peroxydisulfate and iodide is given below:

$$S_2O_8^{2-} + 3I^- \;\langle\!=\!=\!=\!=\!=\!=\rangle\; 2SO_4^{2-} + I_3^-$$

This reaction is kinetically described by the relation:

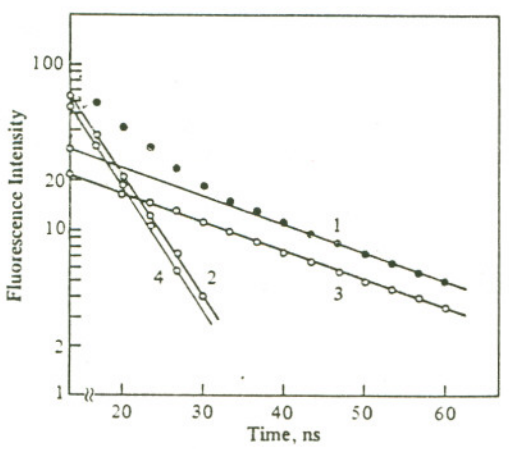
$$d[S_2O_8^{2-}]/dt = -k[S_2O_8^{2-}][I^-]$$

with $k = 2.2 \times 10^{-3}\ M^{-1}\ sec^{-1}$. In order to assess the utility of this reaction scheme, it is necessary to determine the time domain of the reaction. Calculate the time necessary to react 10% of the peroxydisulfate when the solution is 0.01 M in iodide and 5×10^{-4} M in peroxydisulfate. What assumptions are made in the calculation?

$$\int_{x_o}^{x_t} dx/x = \ln(x_t/x_o)$$

2. It has been suggested (Anal. Chim. Acta, 106, 67 (1979)) that excited state lifetime differences can be used to determine two species in a mixture. This is analogous to the simultaneous kinetic determination of two species which react with different rates. From the data below, determine the concentration of each component in the mixture.

Curve 1: mixture; curve 2, 1.5×10^{-5} M 5,7-dibromo-8-quinolinol (pure standard); curve 3, 4.0×10^{-7} M 8-quinolinol (pure standard); curve 4, subtraction of 8-quinolinol contribution from curve 1.



3. When a new enzyme is isolated and purified, its specificity must be determined, particularly if the enzyme is to be used to determine a particular substrate. The specificity can be determined by characterizing k_2 and K_M (Michaelis constant) by measuring initial rates of reaction. For an enzyme following the Michaelis-Menten mechanism as depicted below,

$$E + S \;\langle\!=\!=\!=\!=\!=\!=\rangle\; ES \;\text{-----}\!\rangle\; P + E$$

the rate law is given by

$$d[P]/dt = (k_2[E]_o[S]_t)/(K_M + [S]_t)$$

Initial rates are measured as a function of $[E]_o$. Derive an expression and describe the experimental protocol to obtain k_2 and K_M graphically from the experimental data and the above rate law.